

반도체 칩간 고속 인터페이스의 기초

Gain Kim

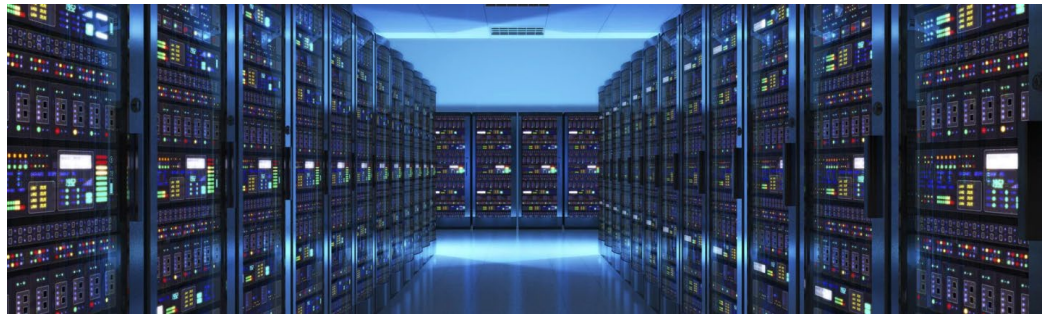
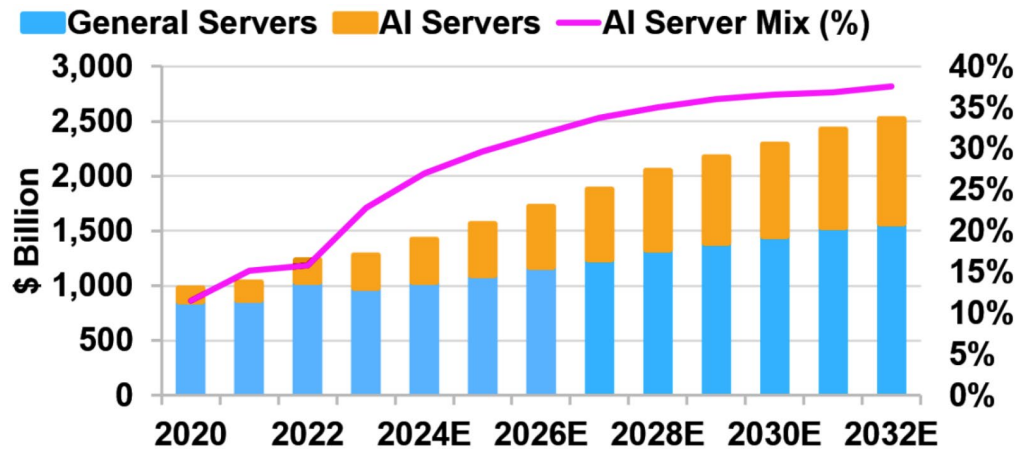
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Chip-to-Chip Interface Applications

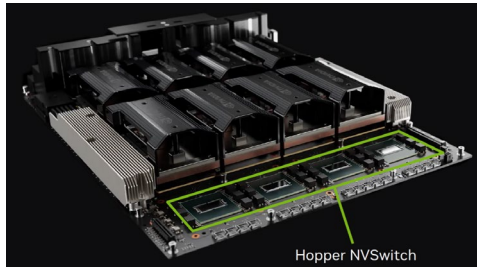
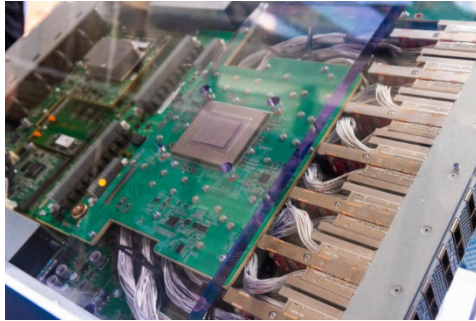
Growth of AI Servers



- ❑ Explosive demand for generative AI fuels the growth of AI servers
- ❑ Generative AI workload growth requires faster networking

Wireline Interconnects in AI Computing System

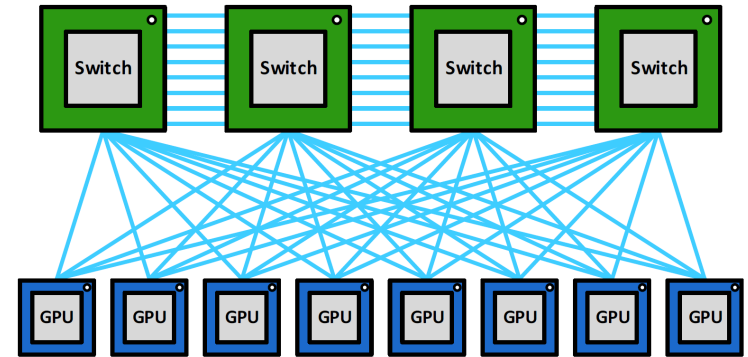
[Celestica]



Hopper NVSwitch

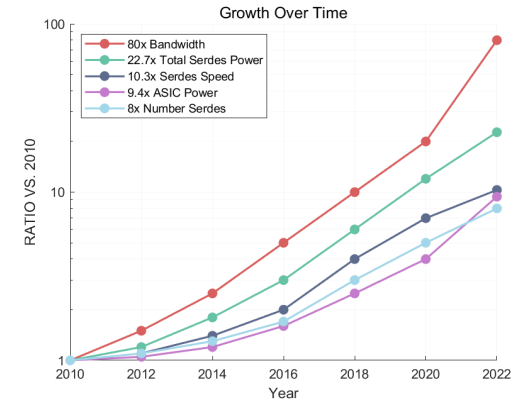
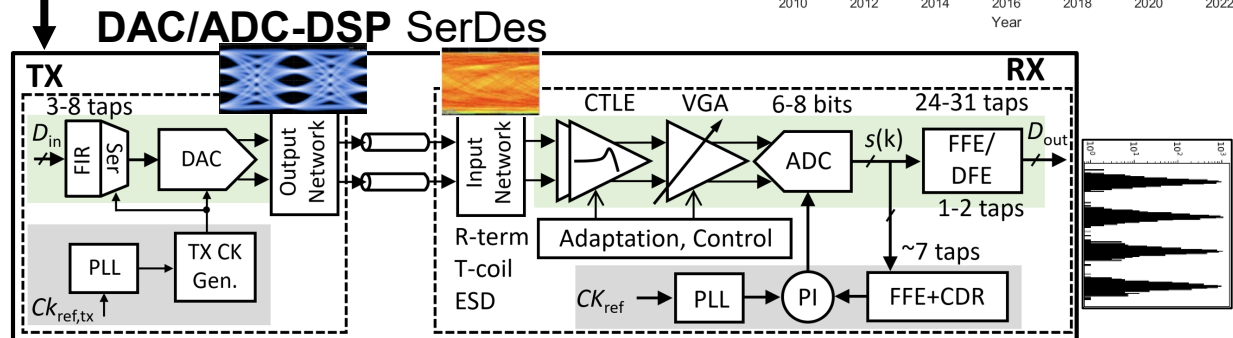
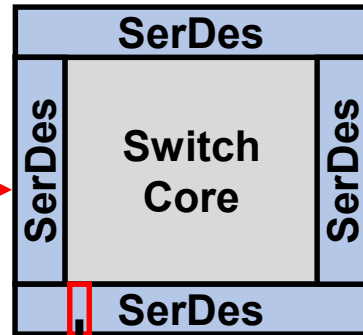
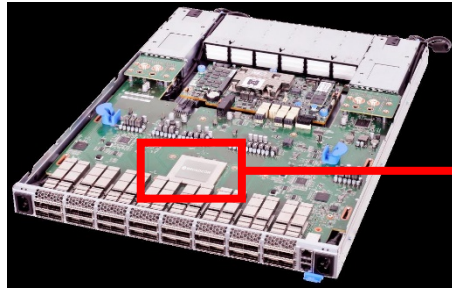
[Nvidia]

2x TOR Switches
3x or 4x Power Shelves
10x 1RU Compute Tray
9x 1RU NVLink Switch Tray
8x 1RU Compute Tray
3x or 4x Power Shelves



- ❑ Modern AI computing systems incorporate multiple compute accelerators on a server, as well as multiple network switches to connect one another
- ❑ Without network switches, processors cannot be interconnected and operate cohesively as a unified computing system

Wireline Transceiver in Network Switch ASIC



❑ >80x switch ASIC total bandwidth

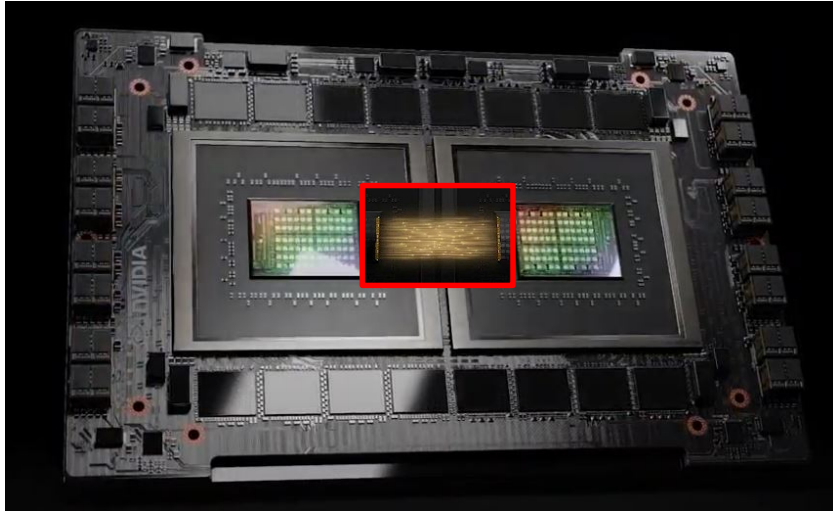
❑ 9.4x switch ASIC power

- 22.7x total SerDes power
- 8x number of SerDes lanes

❑ Per-lane data rate >10.3x increase

Other Applications

Nvidia Grace Hopper CPU

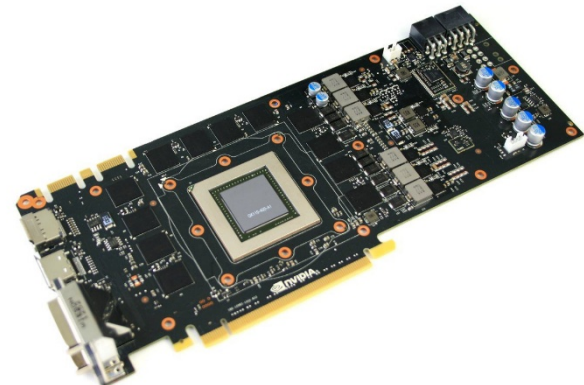


[Nvidia]

Samsung CXL-based memory expander



[Samsung]



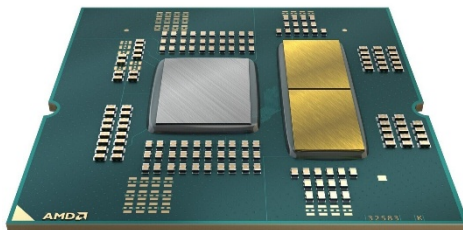
High-Speed Interface Data Rate Trend

❑ Datacenter:

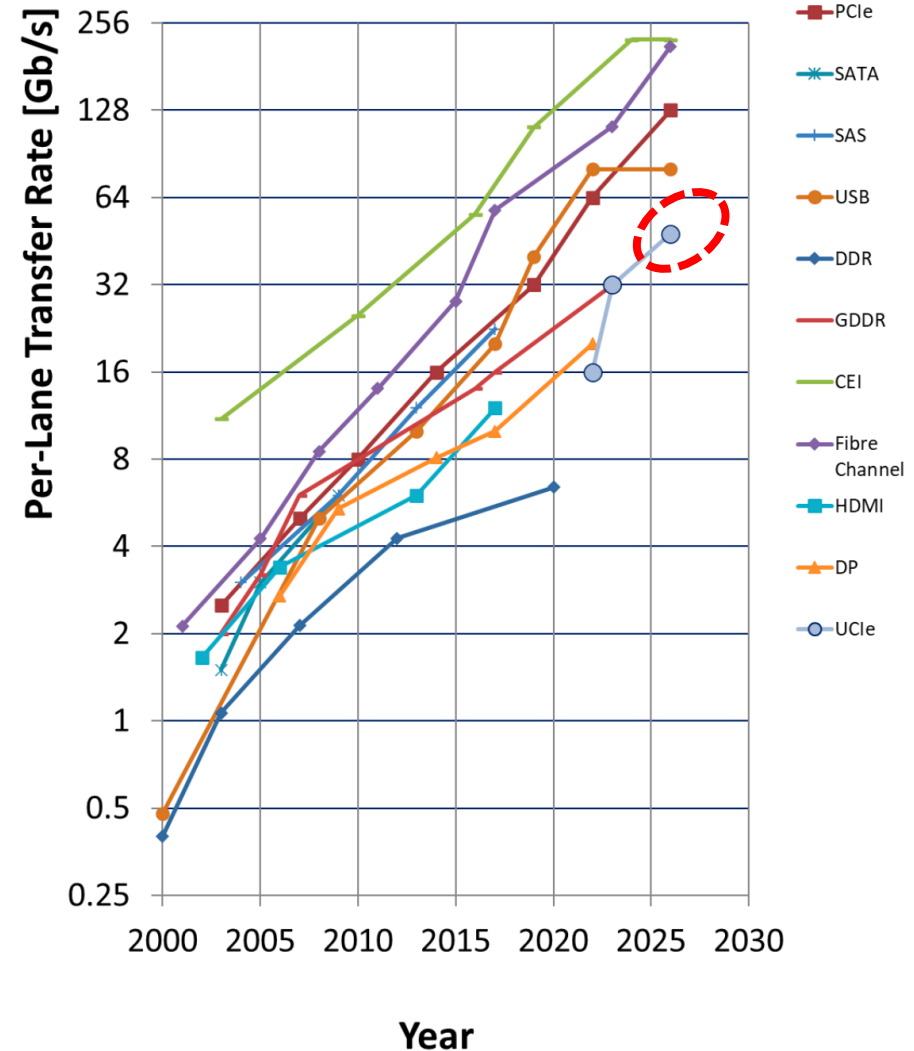
- Long-reach SerDes (ADC-DSP)

❑ AI chiplet / XSR

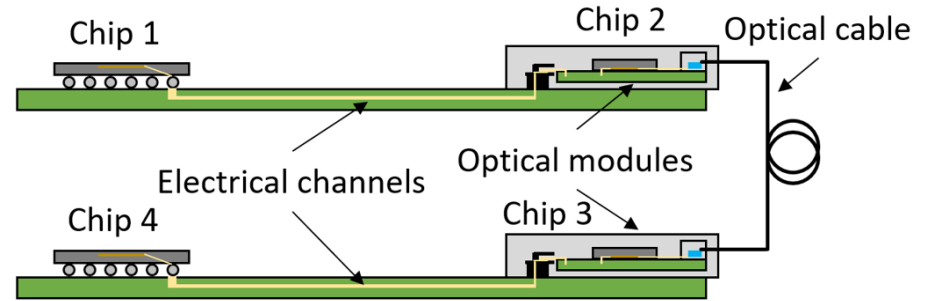
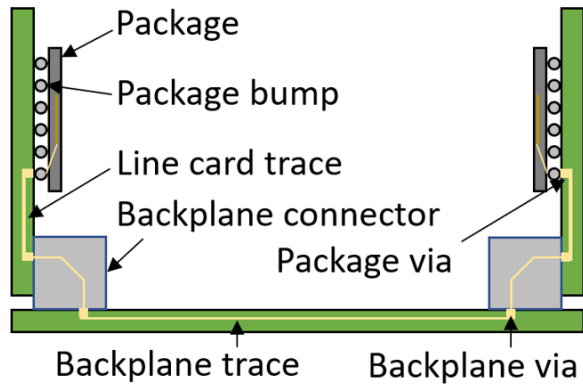
- Analog-intensive
- SBD
- Ultra low-power PHY
- Package-aware design



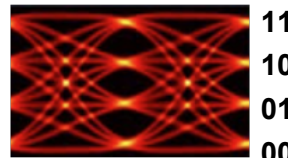
Chiplet



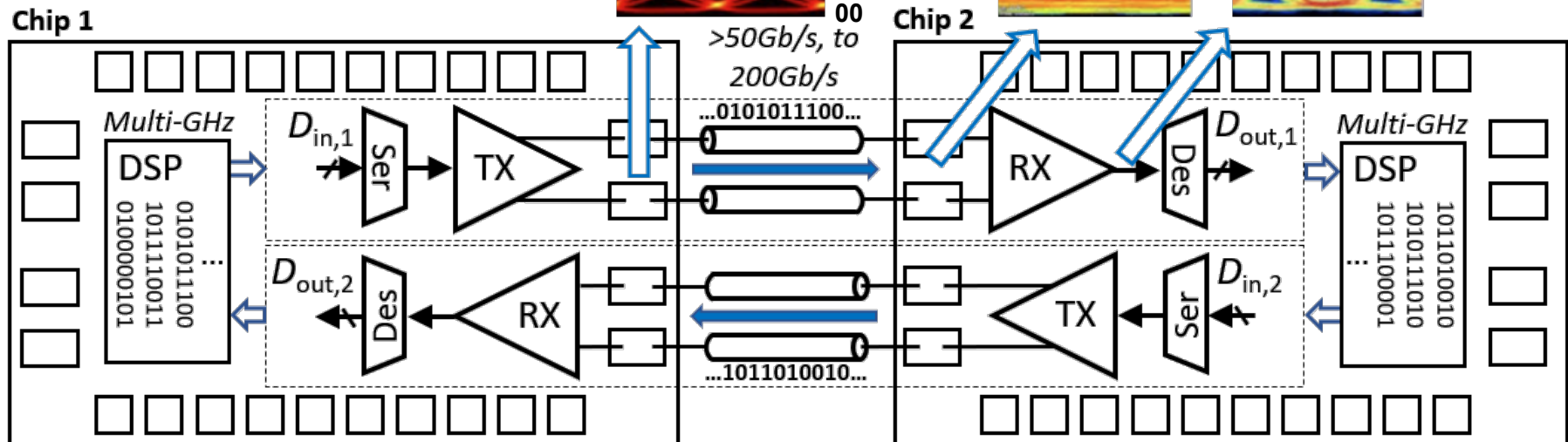
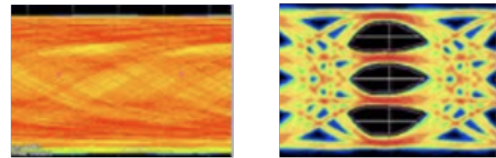
Conceptual View of High-Speed Interfaces



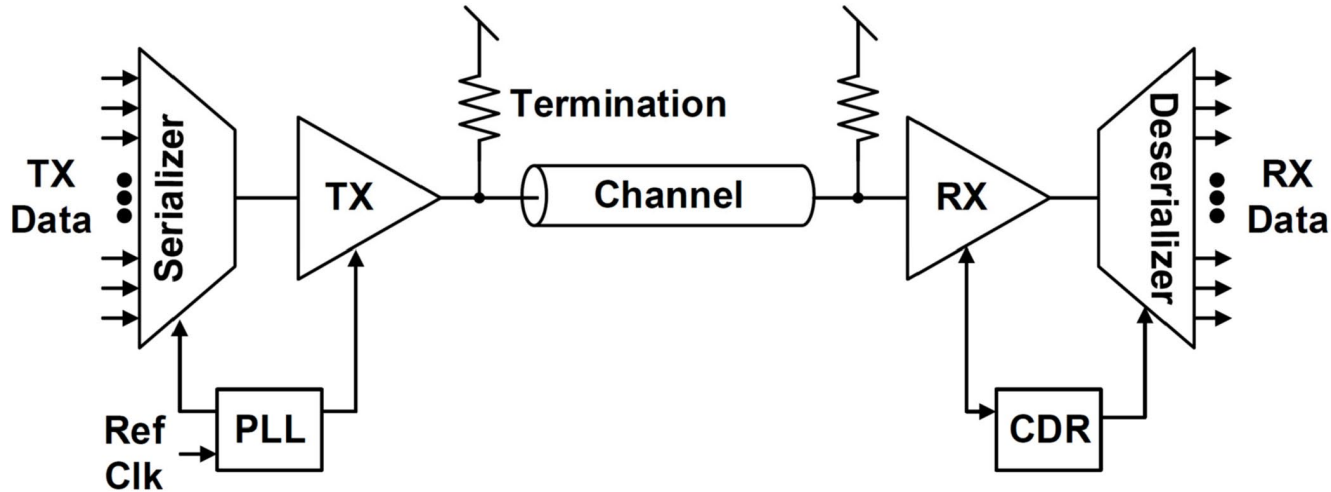
@TX out



@RX in After Equalization



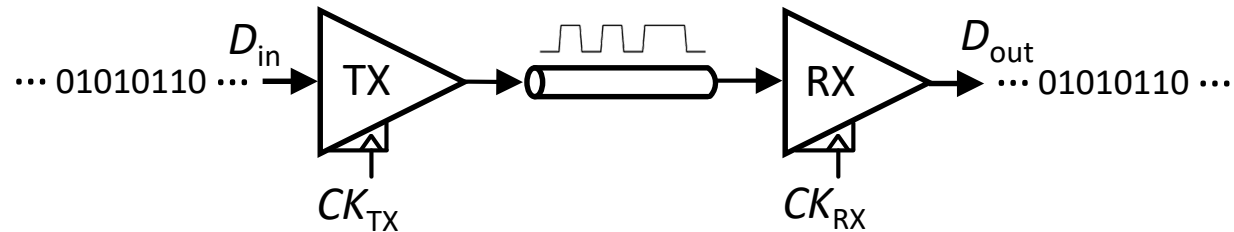
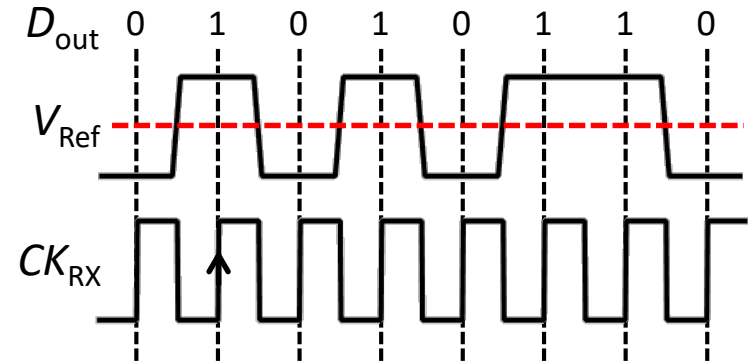
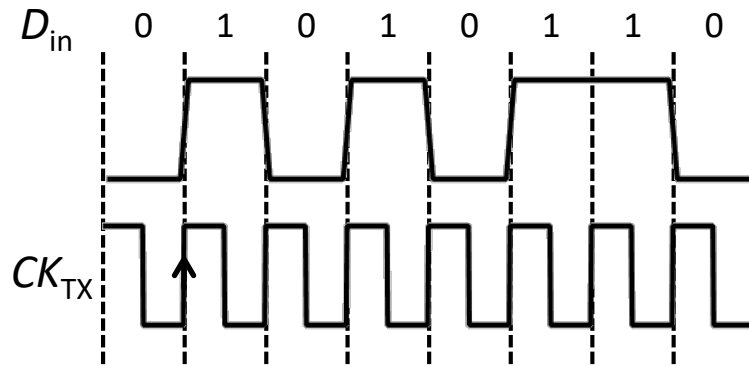
The Need for SerDes



- ❑ Combine multiple data streams and serialize them into a single channel to maximize bandwidth communicated through the limited number of pins (pads)
 - Requires Serializer-Deserializer (SerDes)
 - Channel is transmission line, and complicated TX and RX circuits may be needed

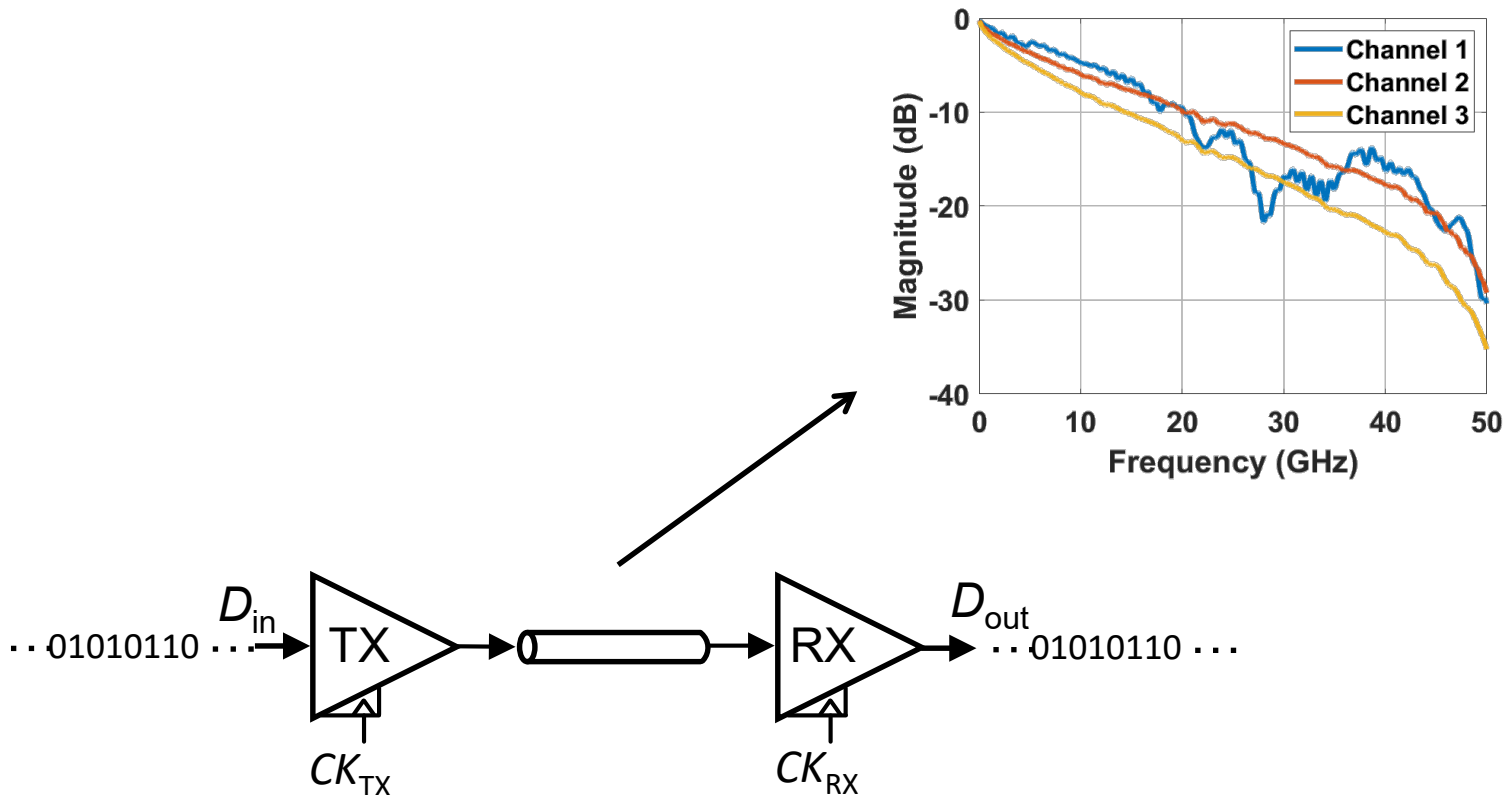
Inter-Symbol Interference (ISI)

Digital Data Transmission and Reception



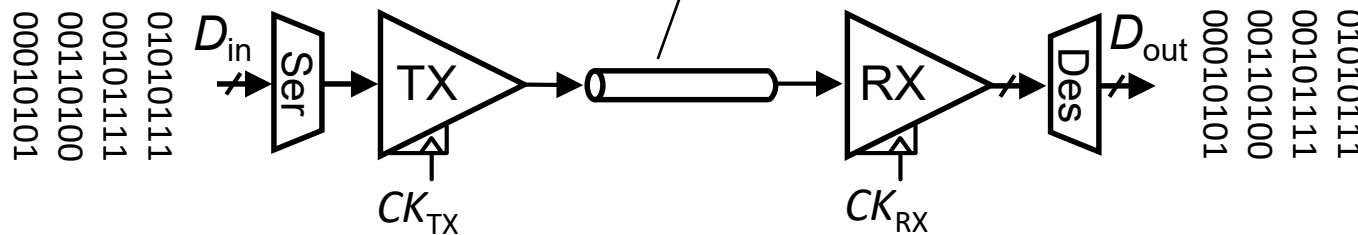
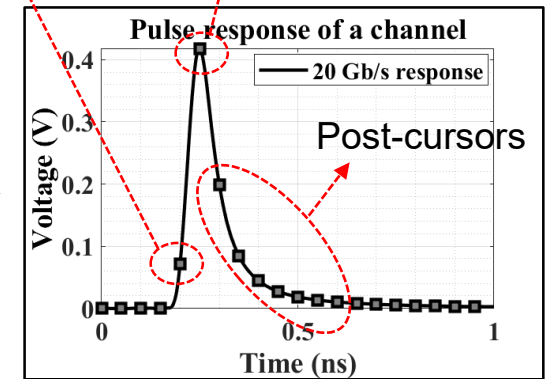
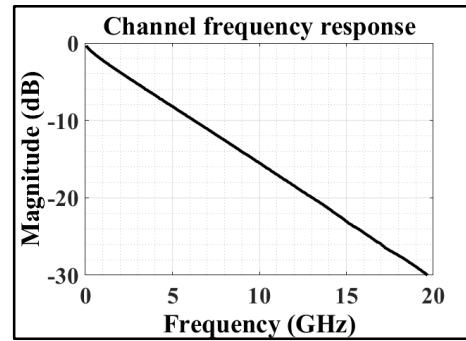
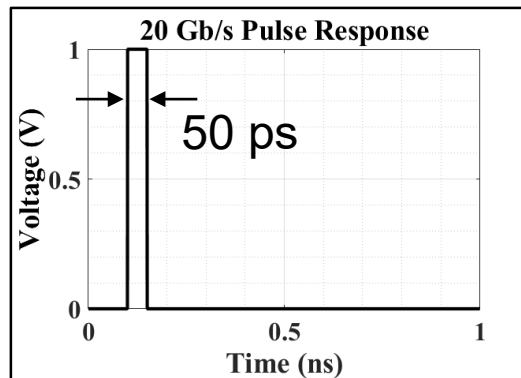
- ❑ Digital data is communicated by sending/receiving binary data
- ❑ At high speed, this simple operation involves many challenges

Channel Frequency Response



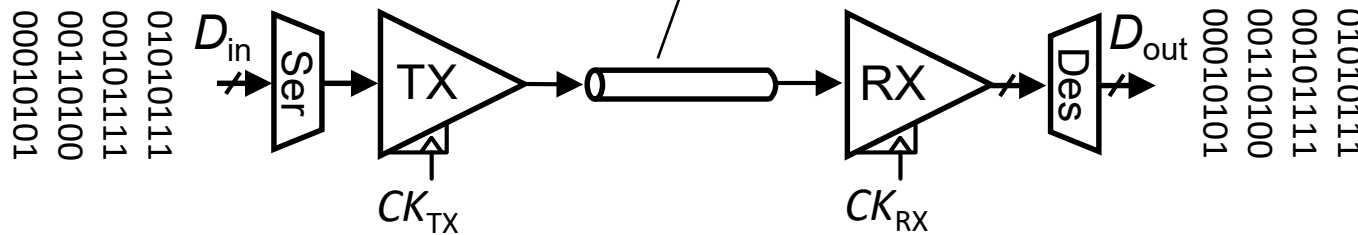
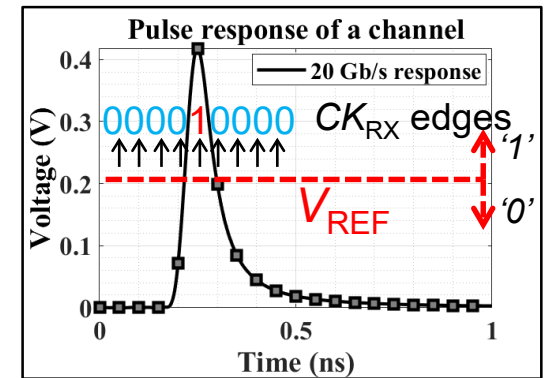
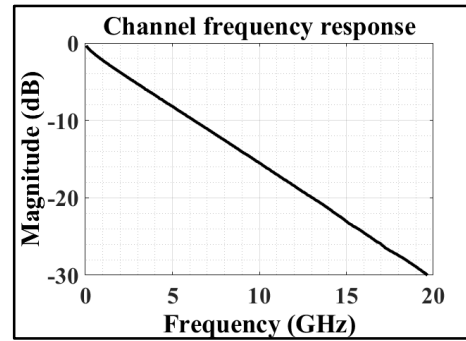
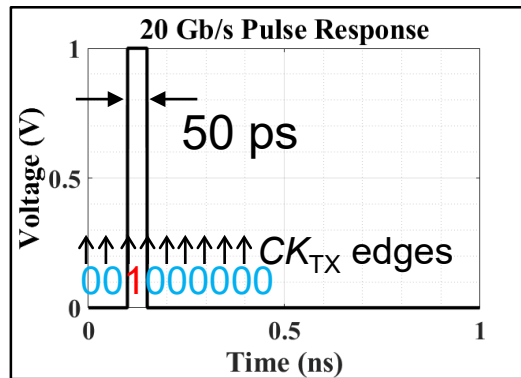
- ❑ Low-pass filter nature (typical point-to-point channels)
- ❑ Linearly increasing loss in dB scale

Channel Pulse Response



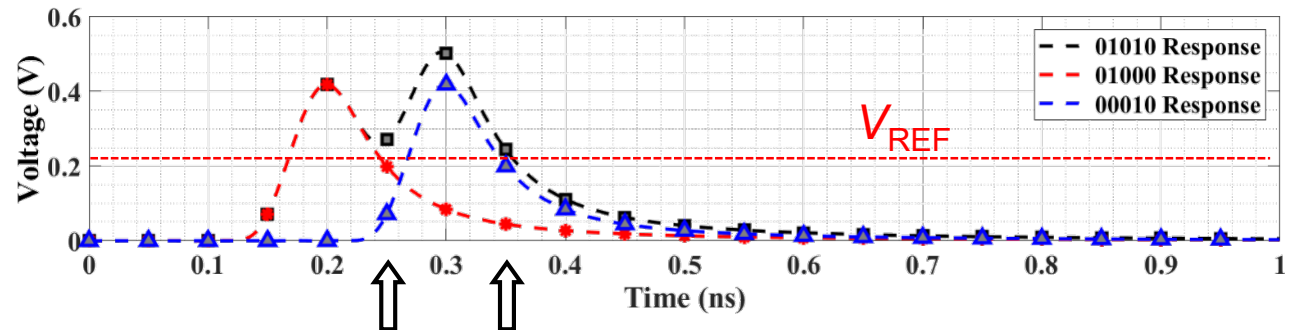
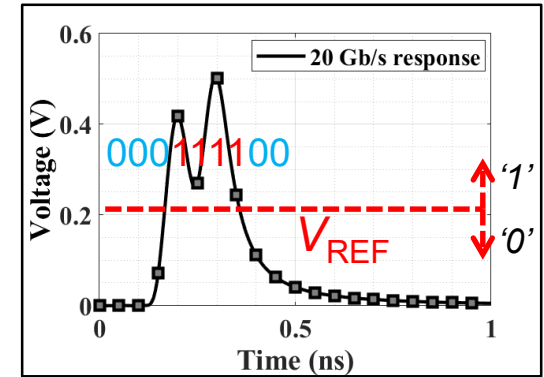
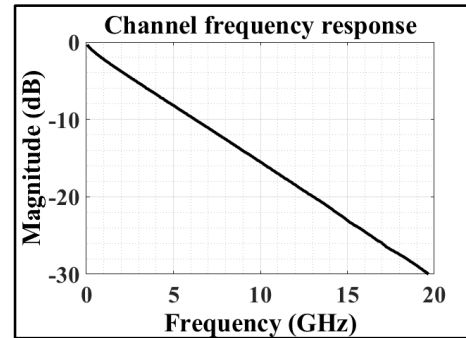
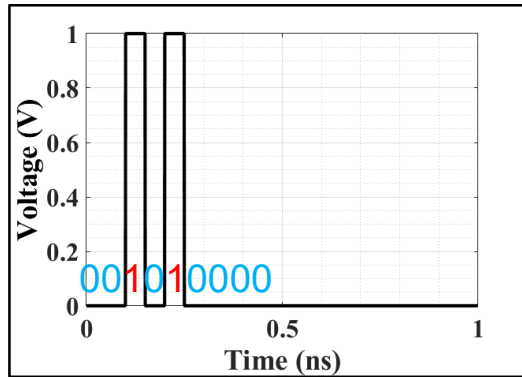
□ Inter-symbol interference (ISI)

Channel Pulse Response



❑ Inter-symbol interference (ISI)

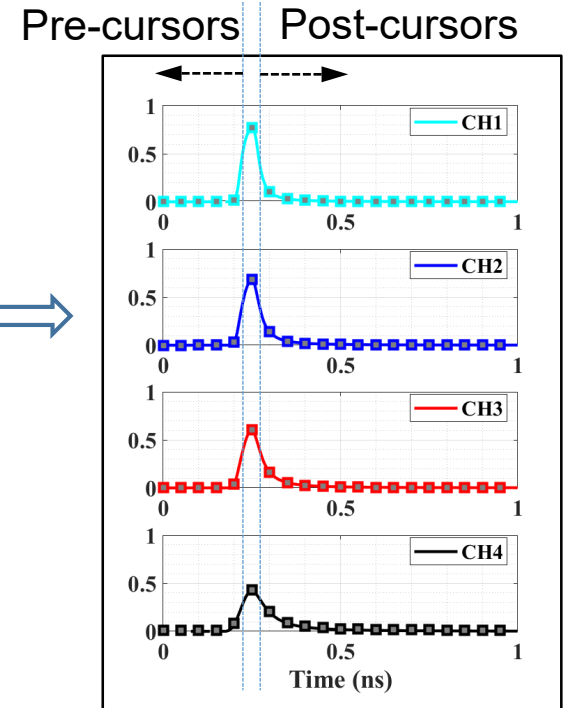
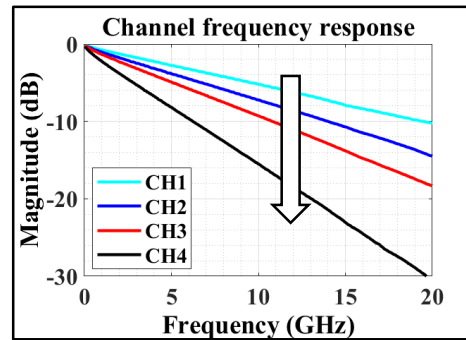
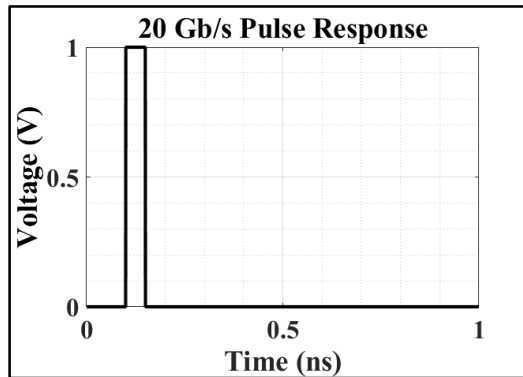
Bit-Error Due to ISI



□ Pre-cursor + post-cursor makes some 0s to 1s

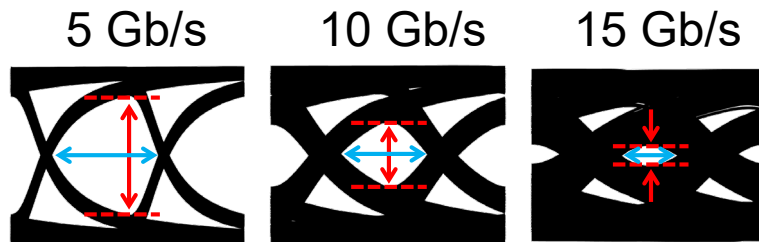
- Bit errors

Channel Loss and ISI Cursors

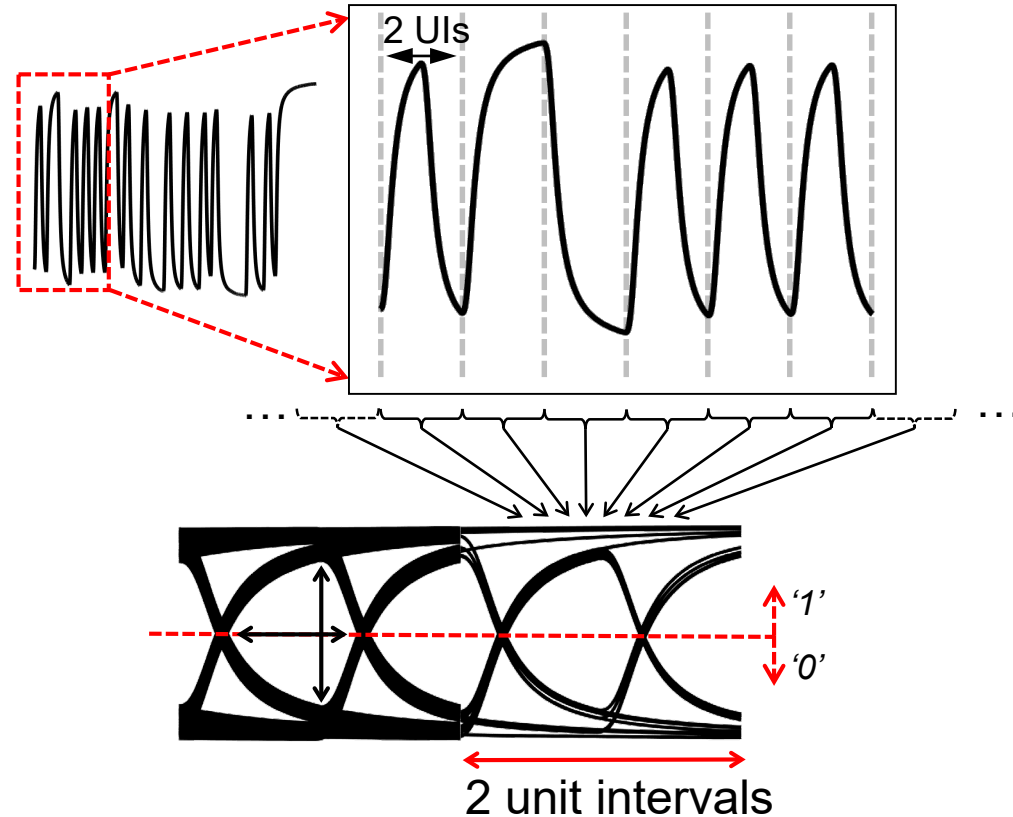


- ❑ Smaller main cursor amplitude for channel with higher loss
- ❑ More and larger pre- and post-cursors for channels with higher loss

ISI and Eye Diagram

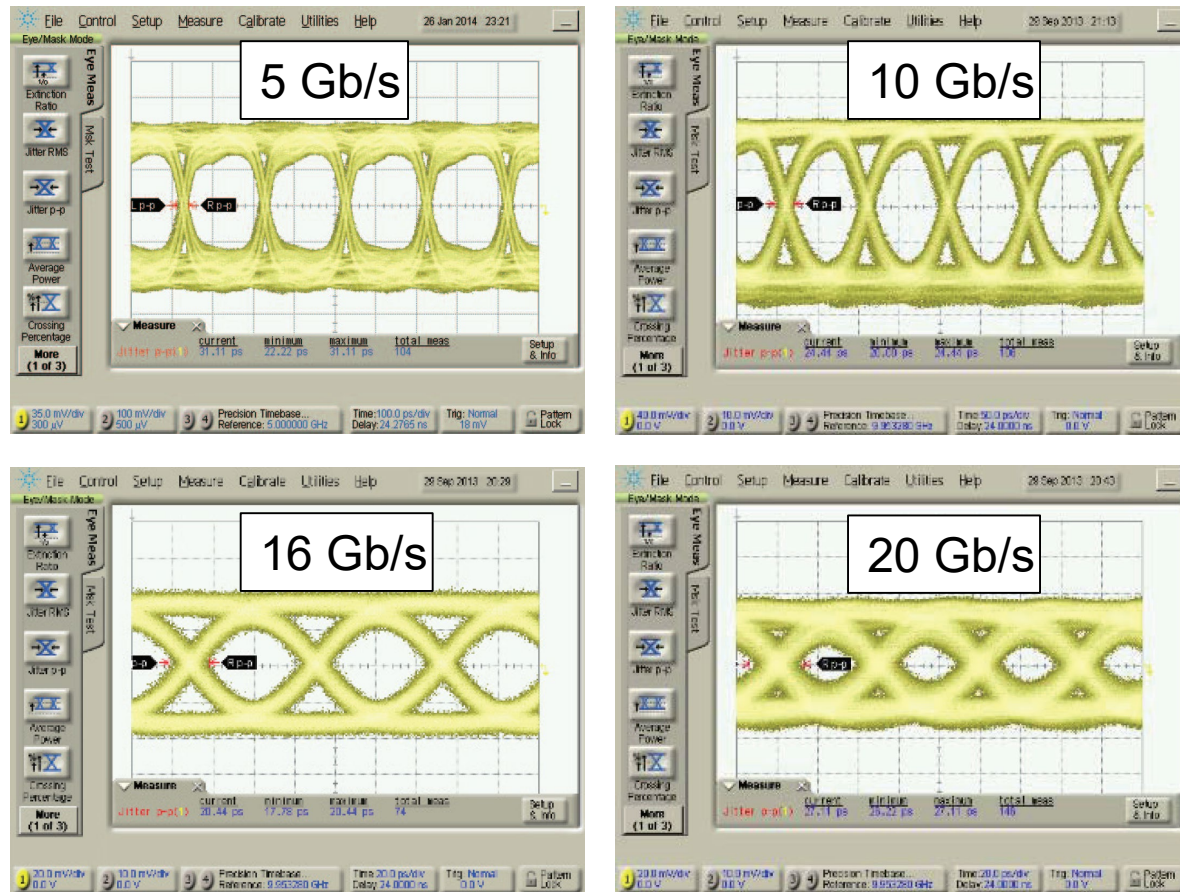


Vertical openings (noise margin)
Horizontal openings (timing margin)



□ Larger ISI makes eye more closed

ISI and Eye Diagram (Example)



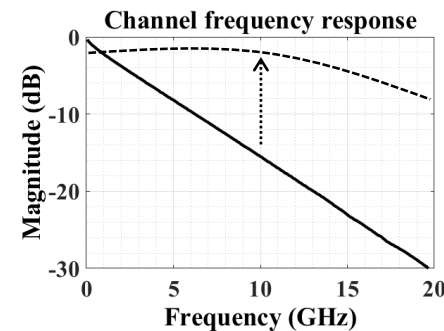
[Y.-F. Lin, ASSCC'14]

❑ Larger ISI makes eye more closed

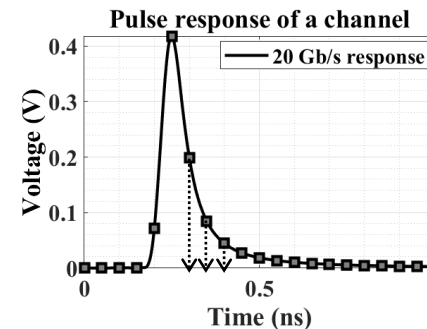
Equalizer (RX CTLE, VGA, DFE)

RX Equalization

- ❑ Make the effective frequency response flat
- ❑ Remove time-domain ISI
- ❑ Linear equalization
 - Continuous-time linear equalizer (CTLE)
 - Variable gain amplifier (VGA)
 - (Feed-forward equalizer (FFE))
- ❑ Non-linear equalization
 - Decision-feedback equalizer (DFE)



Effective freq.
resp. after
EQ.



Zero-force
ISI cursors

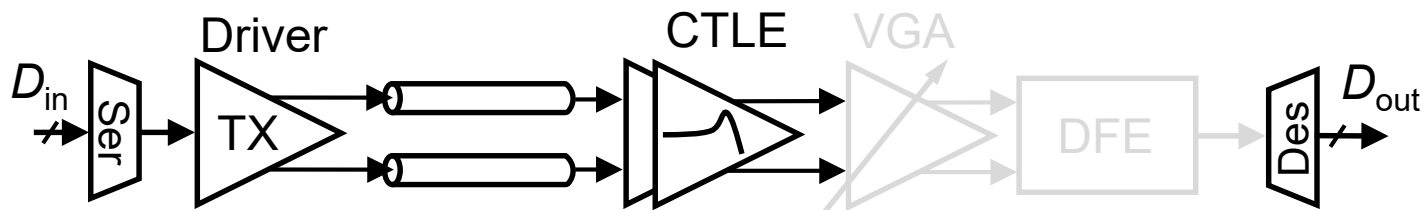
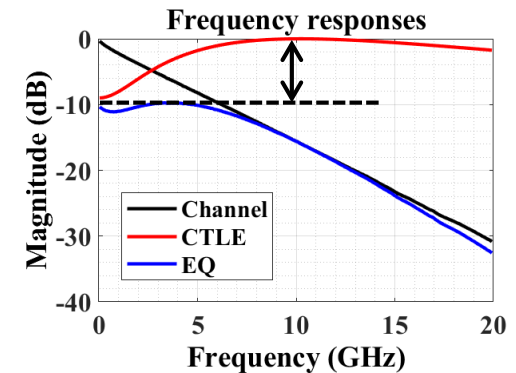
RX CTLE Basics

□ (Partially) compensate for channel loss

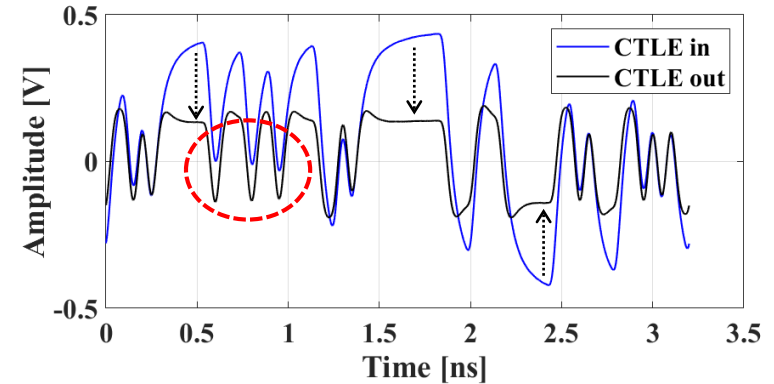
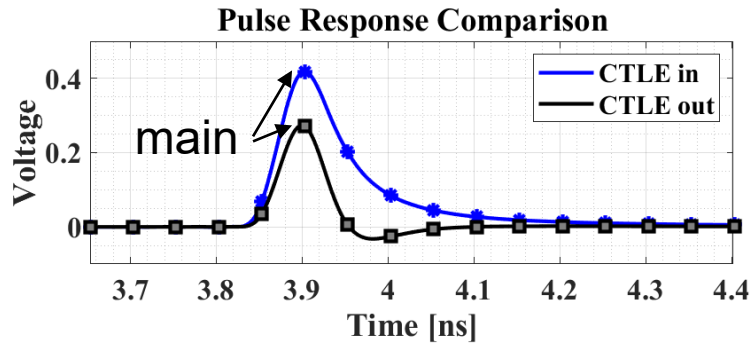
- High-freq. peaking (with low-freq. attenuation)

□ Helps in reducing the amount of low-frequency de-emphasis on the TX FFE

- Increased SNR on the RX input



RX CTLE Basics (2)

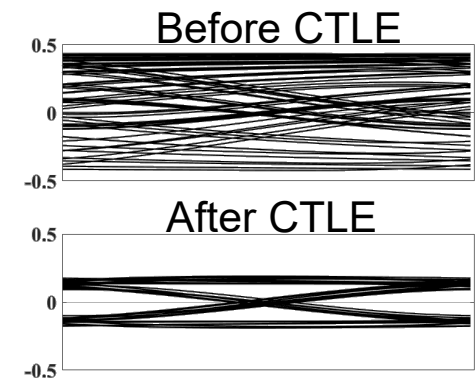


❑ $ISI/main = \sum(abs(cursors_{ISI}))/Peak = 1.17, 0.31$
w/o CTLE w/ CTLE

❑ Shortens the channel pulse response

- Mitigates both pre-cursor and post-cursor ISI

❑ Low-frequency de-emphasis & high-freq. peaking



CTLE+VGA (1)

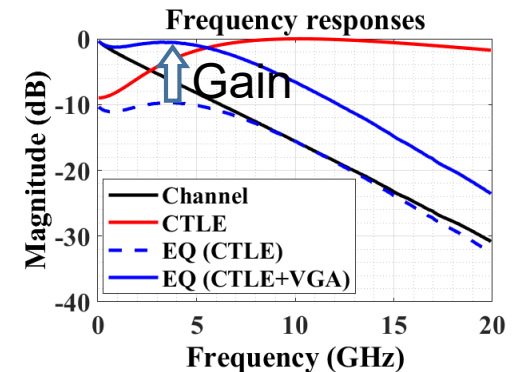
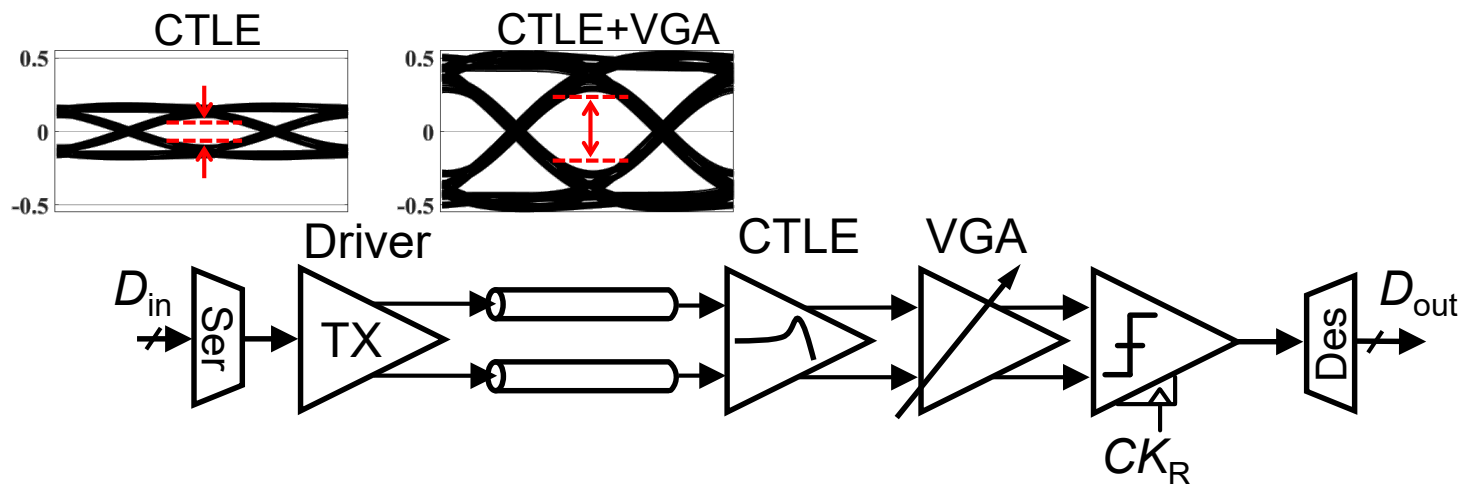
❑ Requires overall signal level adjustment

- Need a variable gain amplifier (VGA)

❑ More margin to the noise on the clocked comparator

- And for the ADC input level adjustment for

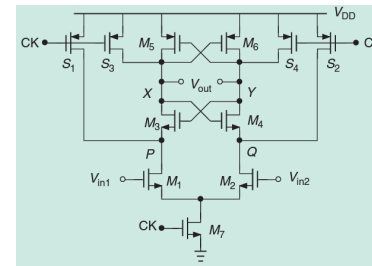
ADC-DSP RX



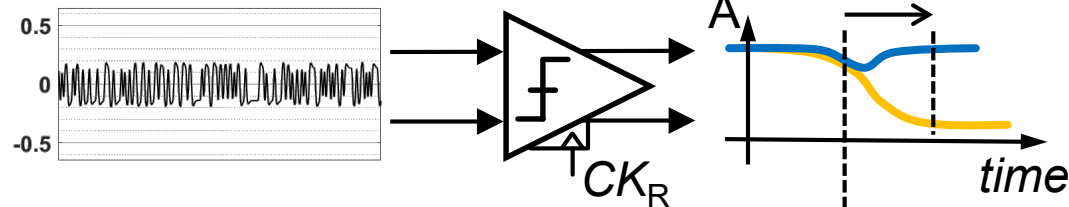
CTLE+VGA (2)

❑ Clocked comparator speed depends on its input swing

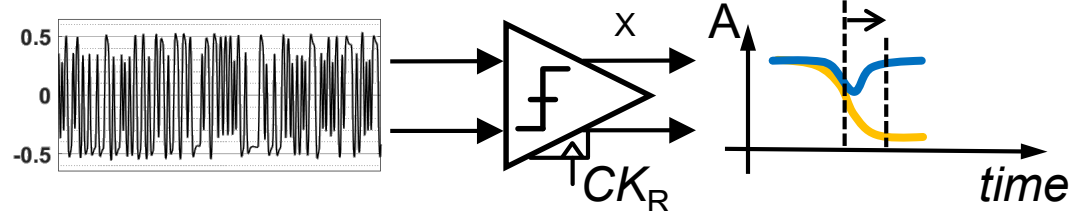
- Regeneration time of the StrongArm latch increases with a small input amplitude
- High input swing is desired



Without
VGA



With
VGA

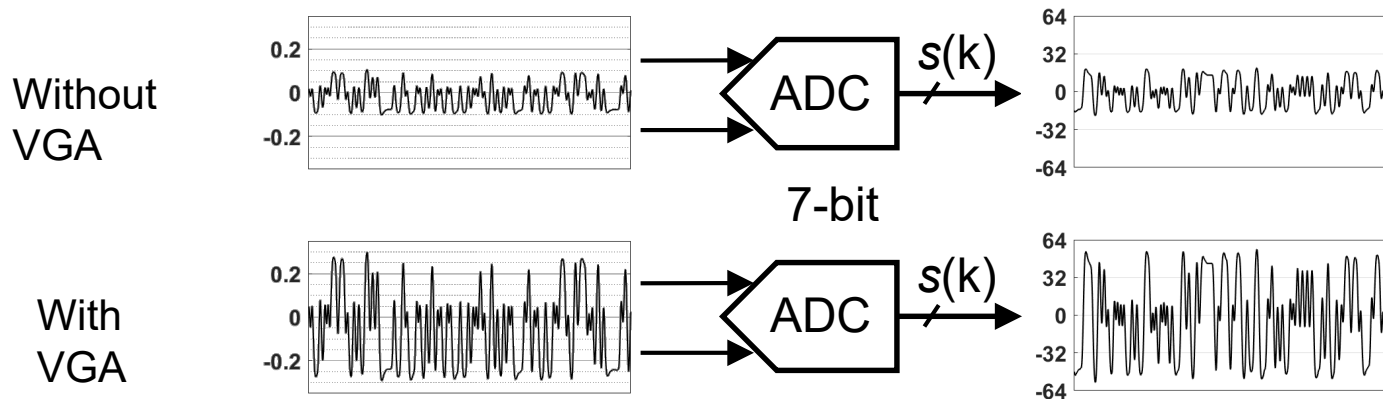


[B. Razavi, "The StrongARM latch," SSCS Magazine'15]

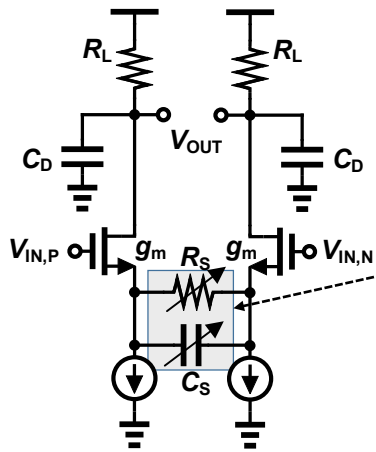
CTLE+VGA (3)

❑ For ADC-based RX

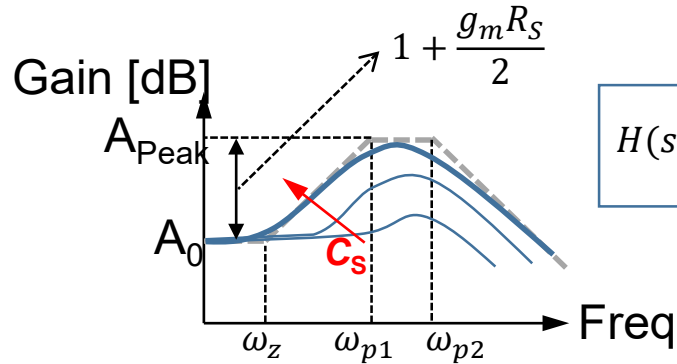
- ADC bits are wasted if the input dynamic range cannot be fully utilized
 - Increased quantization noise
- Adjust the analog input amplitude to the ADC input full-scale dynamic range



CTLE Circuit (1)



Resistive @low freq.
Capacitive @high freq.



$$H(s) = \frac{g_m}{C_D} \frac{s + \omega_z}{(s + \omega_{p1})(s + \omega_{p2})}$$

Gain: $A_{Peak} = g_m R_L$ $A_0 = \frac{g_m R_L}{1 + \frac{g_m R_s}{2}}$

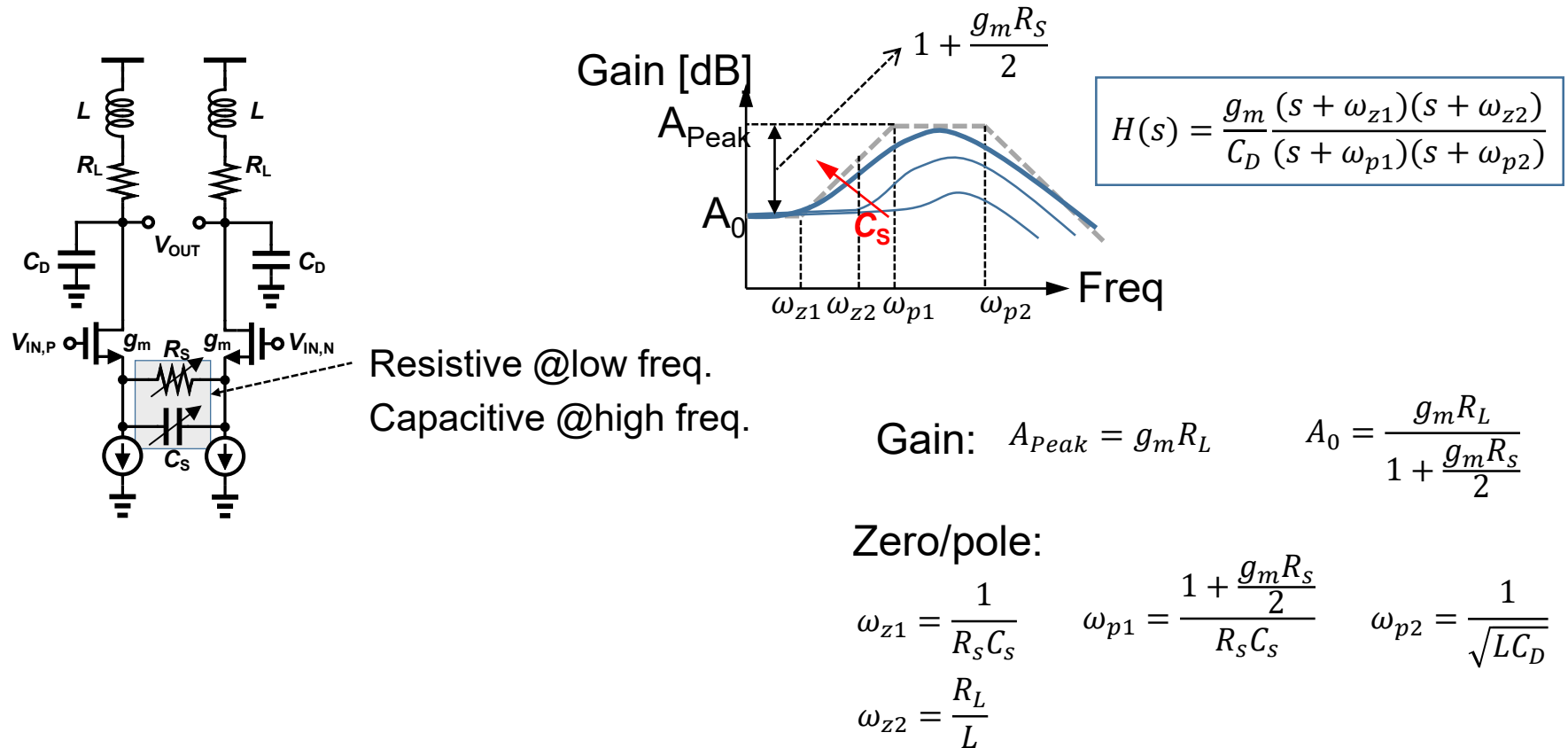
Zero/pole:

$$\omega_z = \frac{1}{R_s C_s} \quad \omega_{p1} = \frac{1 + \frac{g_m R_s}{2}}{R_s C_s} \quad \omega_{p2} = \frac{1}{R_L C_D}$$

□ Source-degenerated differential amplifier

- R, C source degeneration

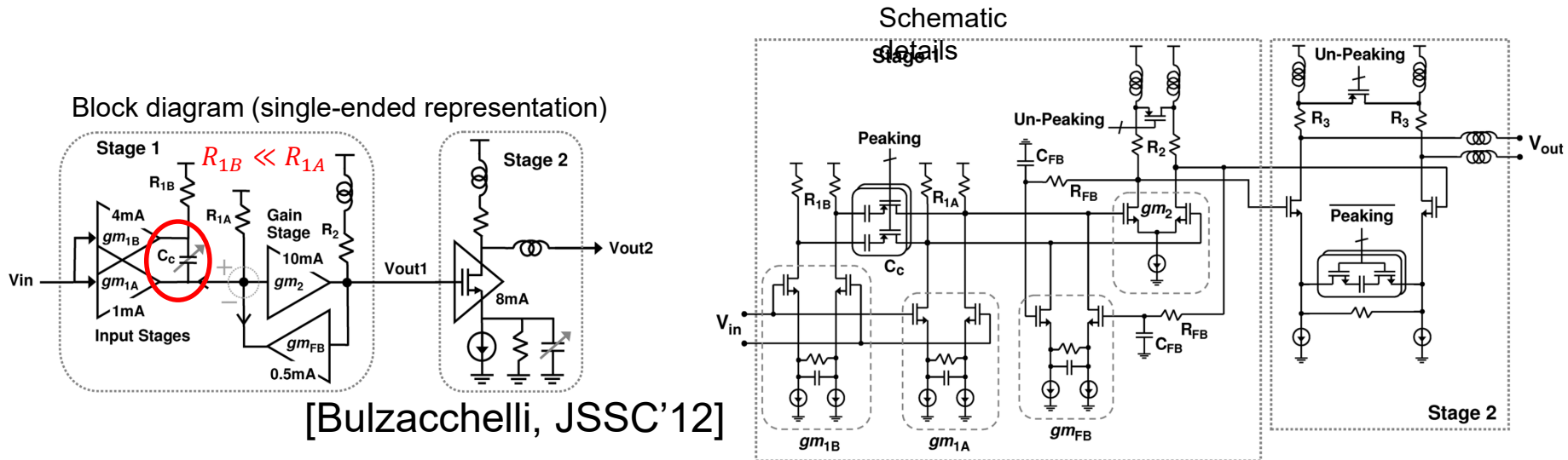
CTLE Circuit (2)



❑ Control R_s and C_s for overall gain, zero, pole

❑ Inductive load for bandwidth extension with additional zero

Alternative CTLE Example



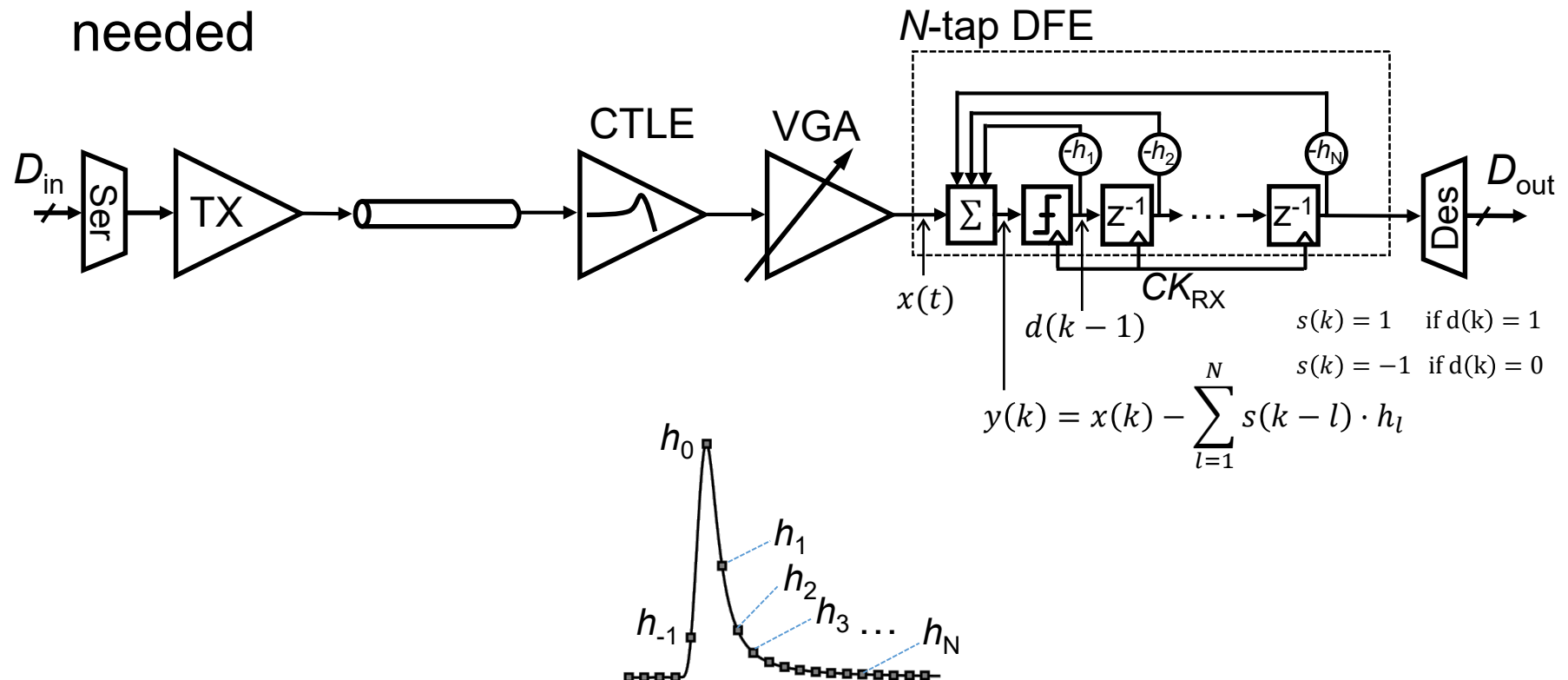
❑ At low frequencies, gm_{1B} is isolated from the rest of the circuit by C_c

$$A_{dc} = \frac{gm_{1A}R_{1A} \cdot gm_2R_2}{1 + gm_{FB}R_{1A} \cdot gm_2R_2}$$

❑ At high freq., $A_{peak,max} = \frac{(gm_{1A} + gm_{1B})(R_{1A} \parallel R_{1B}) \cdot gm_2R_2}{1 + \underbrace{gm_{FB}(R_{1A} \parallel R_{1B}) \cdot gm_2R_2}_{\ll 1}} \approx \underbrace{(gm_{1A} + gm_{1B})(R_{1A} \parallel R_{1B}) \cdot gm_2R_2}_{\text{cascaded two CML stages}}$

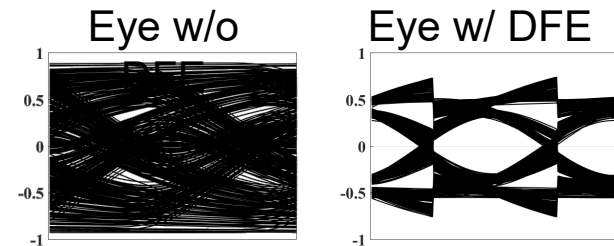
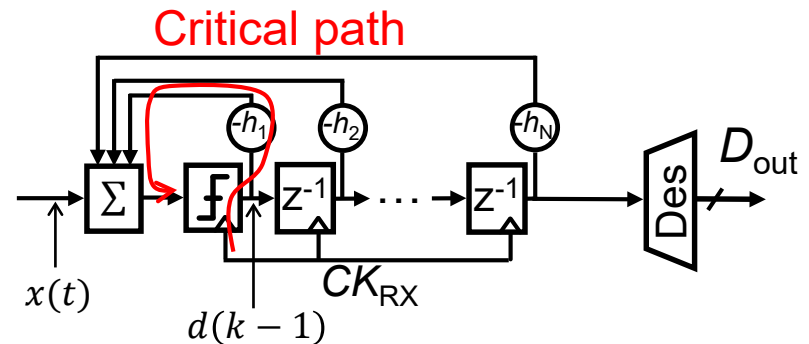
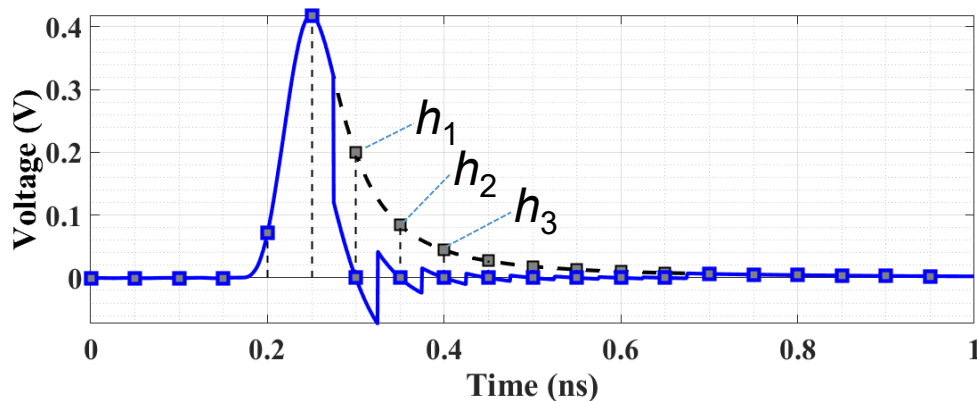
Decision Feedback Equalizer (DFE) (1)

- ❑ DFE can only remove post-cursors
- ❑ As many taps as the number of important post-cursors are needed



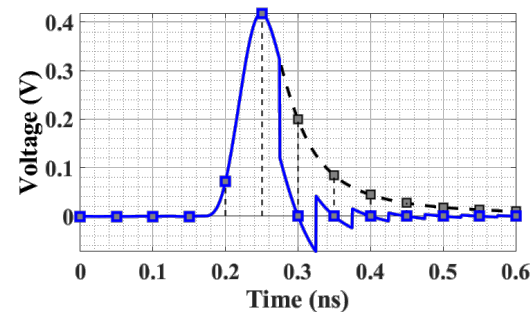
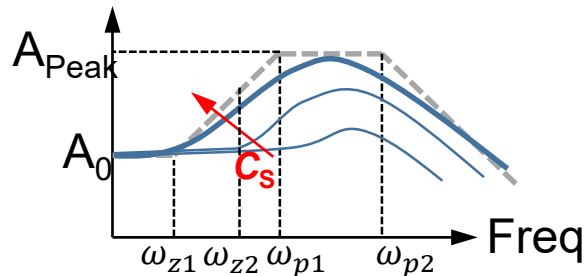
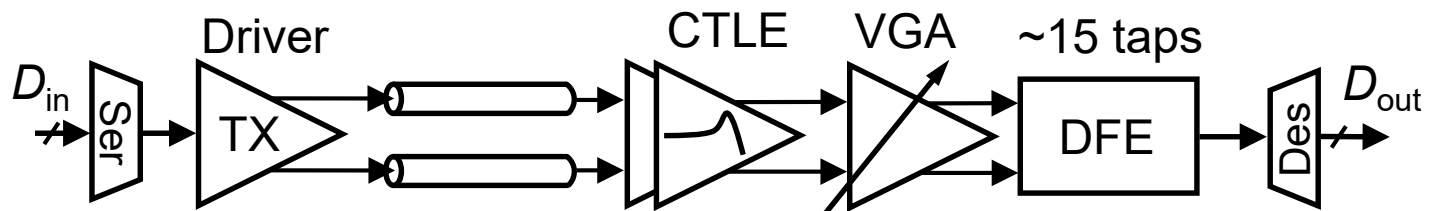
Decision Feedback Equalizer (DFE) (2)

- ❑ Noise and crosstalk are not amplified
- ❑ DFE taps can be adapted without back-channel to TX
- ❑ Symbol-by-symbol feedback requires stringent loop closure timing
- ❑ Pre-cursor cannot be canceled



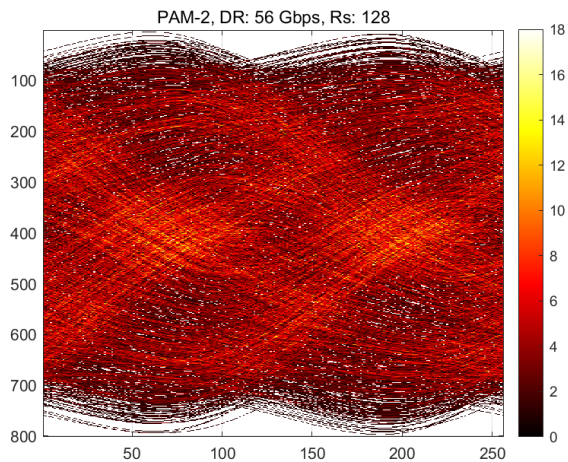
CTLE+VGA+DFE Analog Data-Path

- ❑ CTLE: flatten the frequency response
- ❑ VGA: scale the amplitude level of the signal
- ❑ DFE: time-domain ISI cancellation

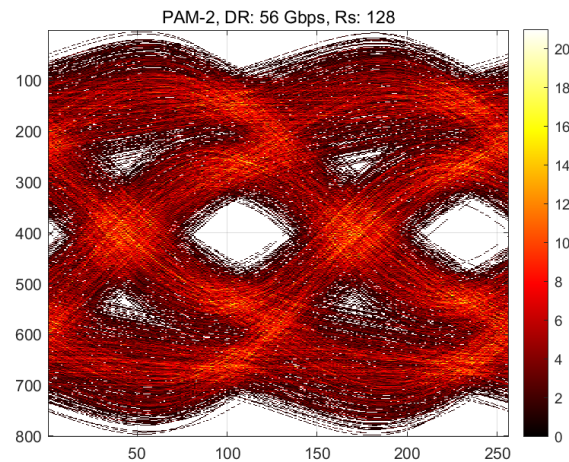


CTLE+VGA+DFE Analog Data-Path

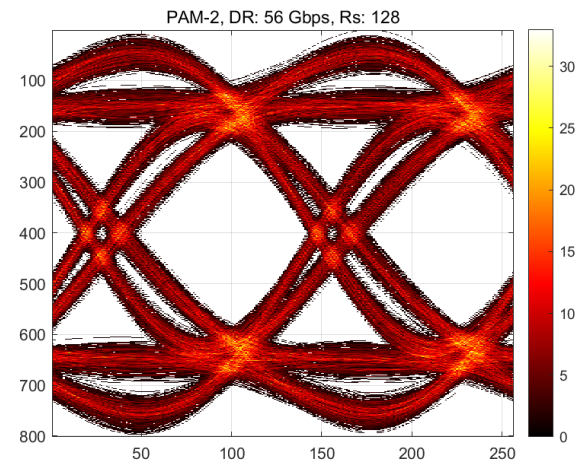
- ❑ CTLE: flatten the frequency response
- ❑ VGA: scale the amplitude level of the signal
- ❑ DFE: time-domain ISI cancellation
- ❑ Equalizer performance in CTLE



Boosting 5dB



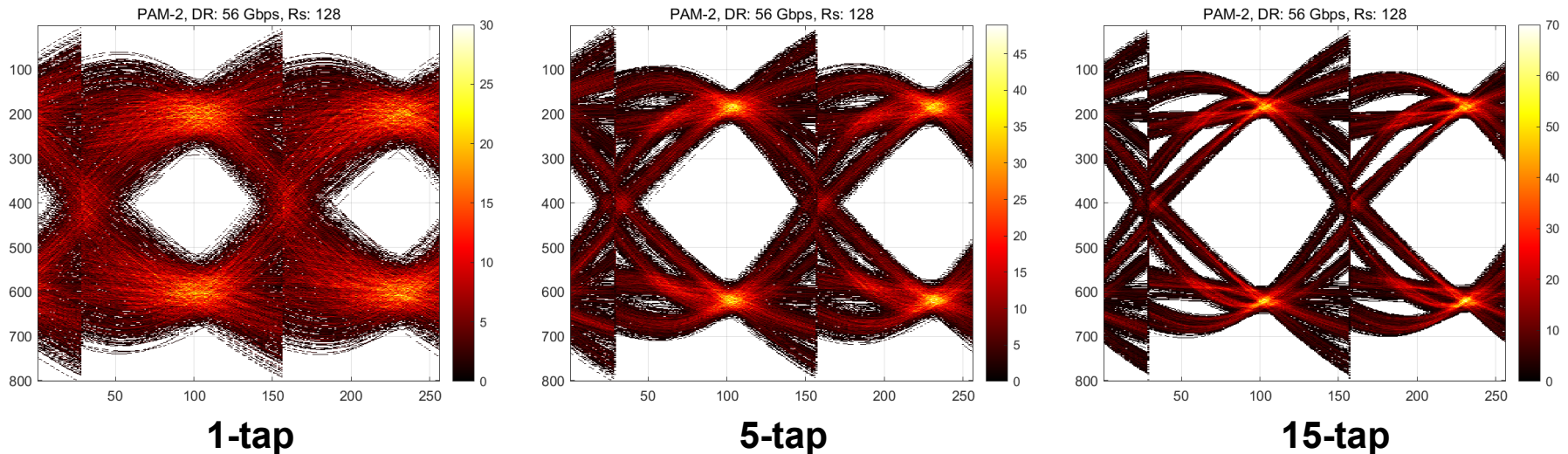
Boosting 10dB



Boosting 15dB

CTLE+VGA+DFE Analog Data-Path

- ❑ CTLE: flatten the frequency response
- ❑ VGA: scale the amplitude level of the signal
- ❑ DFE: time-domain ISI cancellation
- ❑ Equalizer performance in DFE



Summary

- ❑ Channel loss causes ISI.
- ❑ ISI degrades eye opening and BER.
- ❑ TX FFE compensates channel loss.
- ❑ RX CTLE, VGA, and DFE further improve signal quality.
- ❑ Equalization enables reliable high-speed communication.



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